

BANO MARIAM

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in banomariam

🔄 Mariamm97

SUMMARY

Software Engineering graduate with experience in full-stack development, AI/ML, and Computer Vision. Skilled in building intelligent web and mobile applications and computer vision solutions using Python, FastAPI, Next.js, Flutter, and modern AI frameworks. Eager to contribute to software engineering and AI-driven product development.

EDUCATION

COMSATS University Islamabad, Lahore Campus

Bachelor of Science in Software Engineering

2022 - 2026

2nd Place – Final Year Project Exhibition

WORK EXPERIENCE

CacheLogic

Software Engineer Intern

July-2025 - June-2026

Axtra Studio

Software Engineer Intern

February-2025 - May-2025

PROJECTS

Car Inspect – AI-Powered Vehicle Inspection System (Industrial FYP-CacheLogic)

- Developed a smartphone-based computer vision system for automated vehicle inspection.
- Developed a repaint detection module using texture analysis and image enhancement techniques to identify aftermarket paint resprays.
- Built a VIN and chassis authentication pipeline using YOLOv8, OCR ensembles, and image enhancement techniques for accurate VIN extraction across multiple formats.
- Implemented VIN tampering detection using Error Level Analysis (ELA), texture analysis, edge detection, and anomaly detection to identify grinding, restamping, and chassis manipulation.
- Developed a structural damage assessment system using YOLOv8 ensembles for automated detection of dents, scratches, and body damage.

DocuMind AI – Enterprise Document Intelligence Platform

- Built a full-stack RAG-based document intelligence platform for querying PDF, DOCX, TXT, and CSV documents using natural language.
- Implemented a RAG pipeline with document chunking, Sentence Transformer embeddings, and FAISS vector search to retrieve relevant document context.
- Integrated Groq Llama 3 to generate grounded answers based only on retrieved document content, reducing hallucinations.
- Developed a Next.js frontend and FastAPI backend for document upload, indexing, and conversational document search.

SKILLS

- **Programming Languages:** Python, Java, C++, Dart
- **AI/ML:** Deep Learning, Computer Vision, LLMs, RAG, OCR, Semantic Search
- **Libraries:** NumPy, Pandas, OpenCV, Scikit-learn, TensorFlow, YOLOv8, Sentence Transformers, FAISS
- **Framework & Tools:** Django, FastAPI, Flutter, Firebase, REST APIs, GitHub, Docker, Google Colab, VS Code
- **Core Concepts:** Data Preprocessing, Feature Extraction, Model Training, Image Processing

CERTIFICATE

AI Practitioner: Ready to Use and Shape AI – IBM [🔗](#)

Supervised Machine Learning: Regression and Classification [🔗](#)

Python Course-Kaggle [🔗](#)